

ITERATE: Image-Text Enhancement, Retrieval, and Alignment for Transmodal Evolution with LLMs

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Abstract

Inspired by human cognitive behavior, we introduce visual modality to enhance the performance of pure text-based question-answering tasks with the development of multimodal models. However, obtaining corresponding images through manual annotation often entails high costs. Faced with this challenge, an intuitive strategy is to use search engines or use web scraping techniques to automatically obtain relevant image information. However, the images obtained by this strategy may be of low quality and may not match the context of the original task, which could fail to improve or even decrease performance on downstream tasks. In this paper, we propose a novel framework named "ITERATE", aimed at retrieving and optimizing the quality of images to improve the alignment between text and images. Inspired by evolutionary algorithms in reinforcement learning and driven by the synergy of large language models (LLMs) and multimodal models, ITERATE employs a series of strategic actions such as filtering, optimizing, and retrieving to acquire higher quality images, and repeats this process over multiple generations to enhance the quality of the entire image cluster. Our experimental results on the ScienceQA, ARC-Easy, and OpenDataEval datasets also verify the effectiveness of our method, showing improvements of 3.5%, 5%, and 7%, respectively.

1 Introduction

In recent years, with the advancement of large language models (LLMs), the field of Natural Language Processing (NLP) has witnessed unparalleled progress (Floridi and Chiriatti, 2020; Ouyang et al., 2022). From text generation and knowledge-intensive question answering to sentiment analysis, LLMs have demonstrated remarkable performance across a wide range of NLP tasks. However, relying solely on textual information may not be sufficient in various scenarios. For instance, in

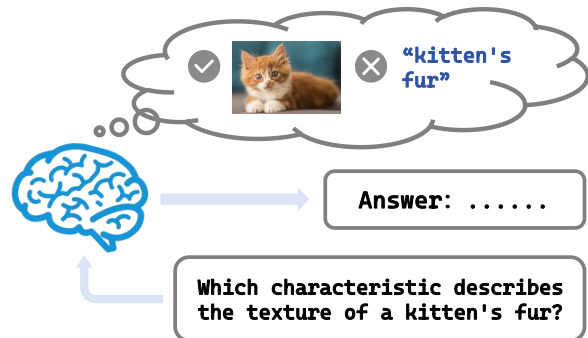


Figure 1: The inspiration for ITERATE. When a person is asked a question, the brain first associates it with the visual information of the relevant content, rather than the natural language itself.

a question-answering task, when the question relates to the essence or characteristics of a subject, merely depending on a text might fall short of capturing the detailed information needed for this task. This raises an interesting question: can we combine other modalities, such as images, with text to allow the model to better comprehend and answer the question? Just as with human cognitive behavior, when humans are asked a question, especially about a specific object, the first thing that comes to mind is the visual information of that object, rather than its natural language description, as shown in Figure 1. By understanding the visual content that is conjured up, one can provide the correct answer. This is one of the inspirations behind the work presented in this paper.

Multimodal question-answering datasets have emerged, such as ScienceQA (Lu et al., 2022), which provides paired images for the questions or options of some examples. However, not all examples in ScienceQA are provided with images, presenting a challenge of missing images. Similarly, for general pure natural language QA tasks, incorporating additional relevant image information can also enhance the model's answering capability.

Inspired by retrieval-enhanced technologies (Gua et al., 2020; Lewis et al., 2020; Izacard and Grave, 2020; Lin and Byrne, 2022; Chen et al., 2022), instead of the traditional method of searching for relevant documents or answers through question texts (or search for related images through a database), we consider a novel approach to align the question text with relevant images to assist the model in answering. However, this strategy brings forth a new challenge: how can we effectively get images that best match the text?

A direct solution is to utilize search engines or web crawlers to obtain images. Yet, there are obvious drawbacks to this approach. Images obtained through search engines or web crawlers can vary in quality, exhibit randomness, and most importantly, may not align with the context of the original task text. Such discrepancies in quality and mismatches with context could lead the model to generate biases during the learning phase, potentially reducing its accuracy and generalization capabilities, hindering the improvement of model performance or even leading to a decrease. How to ensure a high degree of consistency between the image and the text has become our main concern.

Faced with the challenge of selecting the most matched high-quality image for task text, we study different search methods to obtain the image that best matches the text. We find that searching solely based on topic keywords may result in a multitude of mismatches. Therefore, a more refined method is needed to distinguish the internal connections between text and images. Fortunately, Krishna et al. (Krishna et al., 2017) provide invaluable inspiration. They emphasize the profound and intrinsic relationship between natural language and visual content. Natural language can not only identify objects within images but also describe the relationships between them, leading to the proposal of the concept of Visual Genome (Krishna et al., 2017). Using the idea of converting images into combinations of words that describe objects, attributes, and relationships, we can also view images as composed of single or multiple object elements and the relationships between them (ie. transmodal conversion from vision to natural language). This combination of elements and relationships can be regarded as the “DNA” of an image.

In this paper, we draw inspiration from evolutionary algorithm (EAs) (Slowik and Kwasnicka, 2020) and the approach of Krishna et al. to introduce an iterative method named ITERATE. By integrating

the idea of image DNA with EAs, we continuously iterate and refine our image selection, aiming for the utmost consistency between the selected images and their corresponding task texts. Taking advantage of LLMs’ expertise in NLP, we utilize LLMs as a genetic evolutionary operator to generate new “DNA”, with the EAs guiding the entire optimization process. Specifically, based on the initial “DNA” (initial keywords) derived from the task text, we search for related images using a search engine API to create an image cluster. LLMs are employed to imitate evolutionary operators in EAs to generate new “DNA” (new keywords), to search for the next generation of images, and replace those of less quality within the image cluster. We iterate on the updated image cluster to improve its overall quality. To validate the efficacy of the ITERATE method, we implement it on the ScienceQA (Lu et al., 2022), ARC-Easy (Clark et al., 2018) and OpenDataEval dataset. The experimental results indicate that this method not only improves the model’s answering accuracy across the three task sets but also achieves effective performance enhancement on single modal tasks.

2 Related Work

Retrieval Augmented Models For a long time, the combination of image data with text has been a focal point of research, as images encapsulate a wealth of worldly knowledge. Initial research on pre-trained models provided fresh insights for multimodal models. For instance, Flamingo (Alayrac et al., 2022) can generate descriptions from input images. FIBER (Dou et al., 2022) introduced a two-stage visual-language (VL) pre-training strategy suitable for various levels of VL tasks. DALL-E (Ramesh et al., 2021) and Parti (Yu et al., 2022) can generate images based on given text. Blip-2 (Li et al., 2023) initiated language-image pre-training from pre-existing frozen visual and language models. However, these models come with extensive parameter sizes and high pre-training computational costs, and they struggle with unseen knowledge. As a result, many retrieval-augmented approaches emerged to integrate external knowledge from both images and textual documents. For open-domain visual question answering, RA-VQA (Lin and Byrne, 2022) extracts related textual documents from databases through Dense Passage Retrieval (DPR) (Karpukhin et al., 2020) and jointly trains document retrievers and answer generation

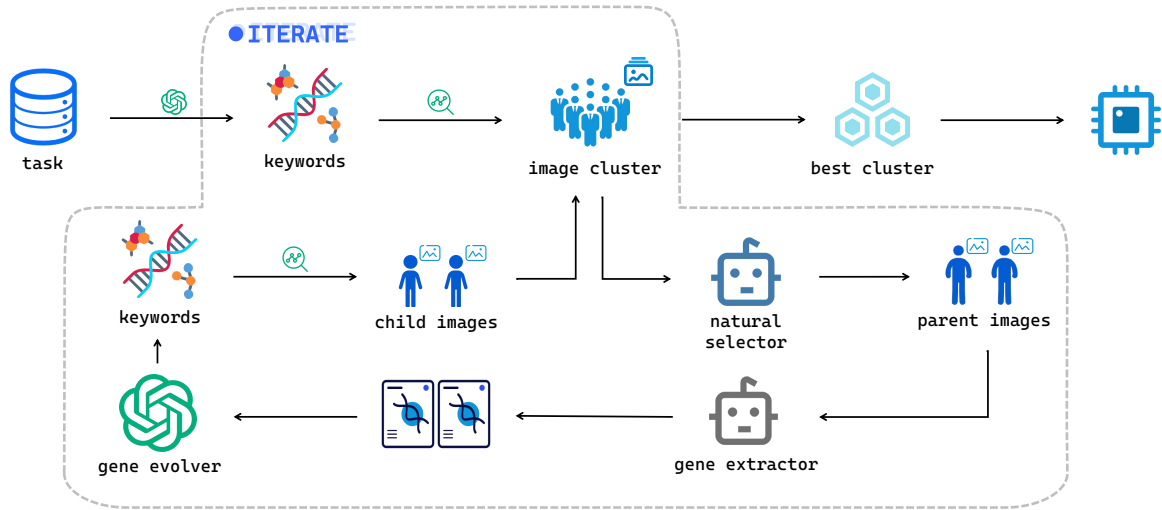


Figure 2: **Flow Chart of ITERATE.** The “natural selector” is the multimodal model used to filter out the parent images for evolution, the “gene extractor” is the multimodal model used to extract the visual content description from the parent images, and the “gene evolver” is the LLM used to execute the evolution operator.

modules. Meanwhile, PICa (Yang et al., 2022) and KAT (Gui et al., 2021) also consider LLMs as implicit knowledge bases. Plug-and-Play (Tiong et al., 2022) utilizes GradCAM (Selvaraju et al., 2017) based on initial questions to locate relevant parts, retrieving associated image patches. Beyond pure text-augmented contexts, MuRAG (Chen et al., 2022) retrieves textual and image data from external storage systems, merging images as visual tokens. In the medical field, RAMM (Yuan et al., 2023) retrieves similar biomedical images and textual descriptions, encoding both modalities through distinct networks. However, these methods’ limitations lie in the need for specific multimodal storage systems for retrieval. Another line of approaches, such as REALM (Guu et al., 2020), RAG (Lewis et al., 2020), and FiD (Izacard and Grave, 2020), integrate Wikipedia articles as data storage, benefiting downstream knowledge-intensive tasks like question answering, but are confined to pure textual knowledge.

Iterative Optimization Method In recent years, significant advancements have been made in the research on iterative optimization methods for LLMs. Yang et al. introduced the “Deep-Thinking” stage (Yang et al., 2023b), which enhances the reasoning abilities of LLMs at test time through the iterative forward optimization of demonstrations. DeepMind proposed the OPRO method (Yang et al., 2023a), leveraging LLMs as optimizers to iteratively generate new solutions based on natural language descriptions and previously discovered so-

lutions for optimization. Iter-RetGen (Shao et al., 2023) enables LLMs to generate natural language reasoning steps or Chains of Thought (CoT) to answer multi-step questions through iterative optimization methods. Some works have also integrated with traditional algorithms in reinforcement learning. PACE (Dong et al., 2023) combines with the Actor-Critic algorithm (Konda and Tsitsiklis, 1999) to realize automatic prompt editing. EVOPROMPT (Guo et al., 2023) is integrated with evolutionary algorithms (Slowik and Kwasnicka, 2020) to achieve iterative optimization of discrete prompts. Promptbreeder (Fernando et al., 2023) is a general-purpose self-referential self-improvement mechanism driven by LLMs, which evolves and adapts prompts based on the given domain. Iterative algorithms have shown their importance in optimization engineering.

3 ITERATE

We consider a question-answering task set $T = \{t_1, t_2, t_3, t_4, \dots, t_{n-1}, t_n\}$. For each sample t , given the input text x , which includes a question q and several options c , the output is a corresponding number (such as A, B, C, etc.) of a selected option. For examples not accompanied by an image, we use the ITERATE method to retrieve the $BestImage = ITERATE(x, n)$ to enhance the model’s capacity to answer the question. Here, $ITERATE(\cdot)$ denotes the iterative evolutionary algorithm, and n represents the number of optimization iterations.

However, the inherent complexity and differences between visual content and textual data pose substantial challenges. For instance, whereas natural language is sequential, images are pixel matrices and inherently non-sequential. This nature makes it challenging to obtain the optimal matching image for a task text. According to the previous work of Guo et al. (Guo et al., 2023), it has been established that evolutionary algorithms are effectively applied for the iterative optimization of phrase sequences in discrete prompts, which are perceived as gene sequences in typical EAs. Utilizing the simplified Visual Genome concept, the visual content of images is transformed into a combination of keywords which is regarded as the image “DNA”, allowing evolutionary algorithms to be employed at the level of natural language.

Following the ideas of Guo et al. (Guo et al., 2023), we leverage the capabilities of LLMs to evolve keyword-based “DNA”. Since the optimization process involves cross-modal conversion, the operations involving the two modalities of image and text are implemented through multimodal models. To achieve the objective of obtaining the best matching image for the task text, evolutionary algorithms have been seamlessly incorporated into the ITERATE method, resulting in the iterative improvement of image quality via the evolution of cross-modal textual “DNA”.

3.1 Framework of ITERATE

Figure 2 depicted the architecture of our ITERATE.

Following the typical EAs, which generally start from an initial cluster and then iterate to generate new individuals by applying evolutionary operators on the current cluster, with each iteration updating the entire cluster, ITERATE primarily encompasses three steps: Initialization, Evolution, and Update:

Initialization: Since there are no initial image clusters corresponding to the samples in the current task set, we need to manually initialize a cluster for each sample. First, LLMs (such as GPT-4) are used to extract the initial keywords from each question text as the DNA of that cluster, represented as

$$keywords = Extr(q, c), \quad (1)$$

where q represents the question text in the sample, and c represents the option text. The search for images is then performed through search engines (such as Google Search, Bing Search), which can

be represented as

$$imgs = search(keywords), \quad (2)$$

to initialize an image cluster, IC . The subsequent iterative optimization process is conducted within this cluster to improve its overall quality.

Evolution: In each iteration, our ITERATE utilizes LLMs as evolutionary operators, which we refer to as the “gene evolver”, and employs multimodal models as cross-modal auxiliary operators, named the “natural selector” and “gene extractor” based on their functions. We use fixed prompts to guide the selection, reverse transcription, and recombinant mutation steps, thereby obtaining new images “DNA” of higher quality.

- **Selection:** Given that ITERATE is an iterative technique involving cross-modal conversion, we simply use the similarity between text and images as a measure of image quality. We employ models (such as CLIP (Radford et al., 2021)) that understand the relationship between images and text to calculate the image-text similarity, represented as

$$similarity = M_{sel}(IC, q), \quad (3)$$

where IC represents the image cluster, and q represents the corresponding question text. Subsequently, the top-2 images in the image cluster are selected as parent images based on the similarity, represented as

$$ParentImgs = top2(similarity). \quad (4)$$

- **Reverse Transcription:** The concept of reverse transcription originates from the relationship between RNA and DNA in biology, where DNA is transcribed into RNA to produce proteins for gene expression, and RNA serves as a template to be reverse-transcribed back into DNA. Considering the combination of natural language keywords as the “DNA” of an image, the process of generating or extracting these natural language descriptions from images can be viewed as a form of “reverse transcription” operation. Just as RNA is reverse-transcribed back into DNA, images are “reverse-transcribed” into the keyword combinations in natural language form. We use multimodal models (such as LLaVA (Liu et al., 2023)) to perform this “reverse transcription” operation, represented as

$$captions = M_{extr}(ParentImgs) \quad (5)$$

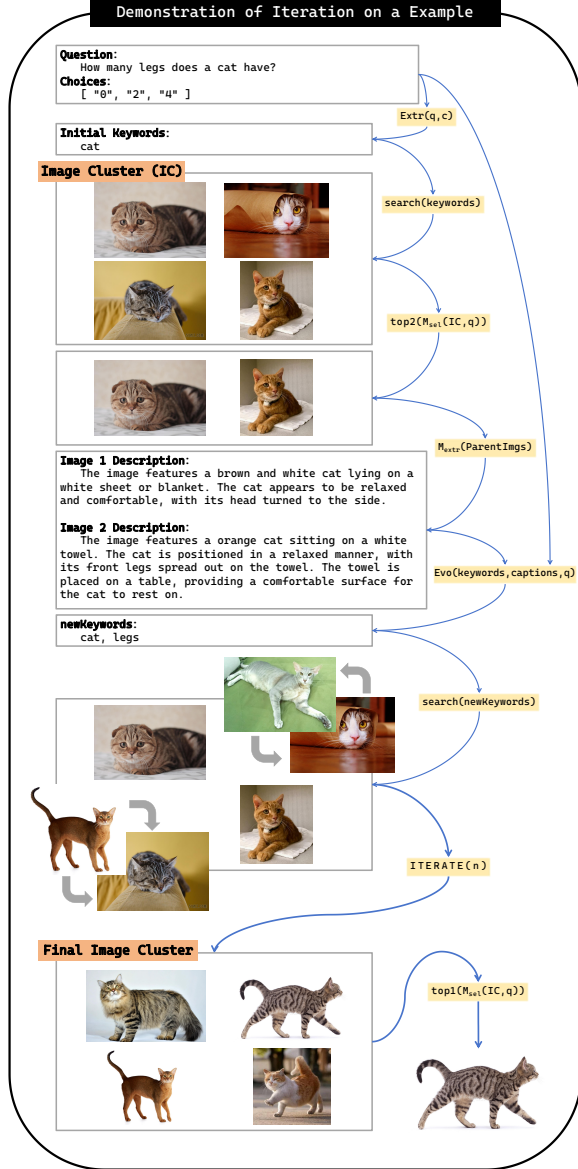


Figure 3: The Iterative Process of ITERATE Algorithm on Demonstration Samples. The operators in the yellow boxes on the right of the figure correspond to Algorithm 1. ITERATE successfully retrieves the best image for the “How many legs does a cat have?” problem.

- **Recombination&Mutation** Based on the two sets of keywords derived from reverse transcription, as well as the task text itself, LLMs are used to “evolve” the original keyword combination into a new set of keyword combinations. The “evolution” operation is akin to integrating the crossover and mutation operations found in evolutionary algorithms: initially, LLMs cross-analyze the commonalities and individualities within the two images’ “DNA”s, and on the basis of retaining common keywords, further select individual ones to construct a new keyword combination to

Dataset	Original	ITERATE	Ratio
ScienceQA	10332	16058	75.7%
ARC-Easy	0	2376	100%
OpenDataEval	0	470	100%

Table 1: Statistical data of examples from the original datasets and after ITERATE image search processing. The “Original” column shows the number of examples with images initially, and the “ITERATE” column shows the count after adding images via ITERATE.

gether with the previous keywords (i.e. **Recombination**); subsequently, **Mutations** are applied to the individual keywords in conjunction with the question text to acquire a new “DNA”. Following these principles, we utilize fixed prompts to guide the behaviors of LLMs, represented by the equation:

$$newKeywords = Evo(keywords, captions, q). \quad (6)$$

Update: ITERATE iteratively generates new keywords DNA and uses search engines to retrieve new offspring images for updating the cluster:

$$ChildImgs = search(newKeywords). \quad (7)$$

We consider a simple and direct updating strategy, specifically, in each iteration, ITERATE searches for N images (set to 2 in this paper) based on the keywords. These new images replace the bottom- N images in the image cluster (those with the lowest text match), resulting in an updated image cluster. This is akin to natural selection, where the fittest survive and produce equally high-quality offspring.

Once ITERATE reaches a pre-set number of iterations, the algorithm stops. Appendix A provides the pseudocode for ITERATE and the fixed prompts we used.

4 Experiments

In this section, we will evaluate our ITERATE method on three datasets. We will then introduce the benchmark dataset, implementation details, performance baselines for comparison, and the final experimental results.

4.1 Dataset

ScienceQA is the first large-scale multimodal multiple-choice science dataset with 21,208 examples across natural science (NAT), social science

Model	Size	NAT	SOC	LAN	TXT	IMG	NO	Avg
MCAN	95M	56.08	46.23	58.09	59.43	51.17	55.40	54.54
Top-Down	70M	59.50	54.33	61.82	62.90	54.88	59.79	59.02
BAN	112M	60.88	46.57	66.64	62.61	52.60	65.51	59.37
DFAF	74M	64.03	48.82	63.55	65.88	54.49	64.11	60.72
ViLT	113M	60.48	63.89	60.27	63.20	61.38	57.00	61.14
Patch-TRM	90M	65.19	46.79	65.55	66.96	55.28	64.95	61.42
VisualBERT	111M	59.33	69.18	61.18	62.71	62.17	58.54	61.87
UnifiedQA _{Base}	223M	68.16	69.18	74.91	63.78	61.38	77.84	70.12
UnifiedQA _{Base} w/ CoT	223M	71.00	76.04	78.91	66.42	66.53	81.81	74.11
GPT-3.5	175B	74.64	69.74	76.00	74.44	67.28	77.42	73.97
GPT-3.5 w/ CoT	175B	75.44	70.87	78.09	74.68	67.43	79.93	75.17
MM-CoT _{Base}	223M	82.24	83.13	82.64	80.84	76.15	85.64	82.53
MM-CoT _{Base} + ITERATE	223M	83.88	92.91	84.45	82.65	82.05	86.90	85.92

Table 2: **The Results of Model Performance Comparison on ScienceQA (%)**. Size = backbone model size. Question categories: NAT = natural science, SOC = social science, LAN = language science, TXT = with text context, IMG = with image context, NO = with no context.

(SOC), and language science (LAN). The dataset is split into training, validation, and testing sets in a 60:20:20 ratio, with 12,726, 4,241, and 4,241 examples, respectively (Lu et al., 2022).

ARC-Easy is a multiple-choice dataset from U.S. middle and elementary school science exams, covering biology, chemistry, physics, earth science, and astronomy.

OpenDataEval is a multiple-choice dataset created by retrieving random articles from the HuggingFace Wikipedia English dataset, constructing chains of thought, and using large language models to score and filter the questions.

4.2 Baseline Models

For the ARC-Easy and OpenDataEval datasets, which are pure NLP tasks, we utilize models from the Llama-2 family and the base model of LLaVA-13B, with the ARC-Easy dataset also incorporating the Mistral-7B (Jiang et al., 2023) model.

As for ScienceQA dataset, following the work of Lu et al., we compare our ITERATE method with the baselines of the following four methods:

- (1) Visual question answering models: MCAN (Yu et al., 2019), Top-Down (Anderson et al., 2018), BAN (Kim et al., 2018), DFAF (Gao et al., 2019), ViLT (Kim et al., 2021), Patch-TRM (Lu et al., 2021), VisualBERT (Li et al., 2019).

- (2) Text-to-text LMs: UnifiedQA (Khashabi et al., 2020), as well as UnifiedQA with Chain-of-Thought (CoT) (Lu et al., 2022).

- (3) GPT-3.5 models: GPT-3.5 based on the text-davinci-002 engine (Chen et al., 2020), as well as GPT-3.5 with CoT (Lu et al., 2022).

- (4) Multimodal-CoT_{Base}: MM-CoT running under the same parameters as our experiments.

4.3 Implementation Details

For MM-CoT, we use GPT-4 for initial keyword extraction and evolutionary operations. Clip-vit-base-patch32 (Radford et al., 2021) is employed for image-text matching, and LLaVA-Lightning-MPT-7B (Liu et al., 2023) for extracting natural language descriptions from images. Bing Search v7, comparable in accuracy to Google, is used as the search engine.

For LLaVA, its multimodal capabilities allow it to handle keyword extraction, evolutionary operations, and natural language descriptions of parent images independently, without external models.

To ensure a fair comparison, ITERATE optimization on ScienceQA only searched for images for examples without them, leaving existing images unchanged. Since few images match the language science category, ITERATE was not applied there. For ARC-Easy and OpenDataEval, images were supplemented for every question. Figure 3 illustrates ITERATE applied to the demonstration.

Method	ARC-Easy
Mistral-7B	80.0
LLaMA-7B	72.8
LLaMA-13B	78.9
LLaMA-33B	80.0
LLaVA-13B	78.0
LLaVA-13B + ITERATE	82.8

Table 3: The Results of Model Performance Comparison on ARC-Easy (%).

Method	OpenDataEval
Mistral-7B	66.7
LLaMA-7B	60.1
LLaMA-13B	66.5
LLaVA-13B	64.8
LLaVA-13B + ITERATE	72.0

Table 4: The Results of Model Performance Comparison on OpenDataEval (%).

For the ScienceQA dataset, we fine-tuned the Multimodal-CoT model (Zhang et al., 2023) to evaluate our ITERATE method. DETR (Carion et al., 2020) generated visual features, and FLan-Alpaca-Base (Chia et al., 2023) served as the backbone. The models were fine-tuned for up to 20 epochs with a learning rate of $5e^{-5}$, a maximum input length of 512, and a batch size of 8.

For the ARC-Easy and OpenDataEval datasets, we adopt zero-shot learning on the LLaVA-v1.5-13B model to test the robustness of our ITERATE method in generalization and the helpfulness of image modal in NLP question-answering tasks.

4.4 Original vs. Dataset after ITERATE

We also performed statistics on all examples in the three datasets enhanced by the ITERATE algorithm, as shown in Table 1.

For ScienceQA, the original dataset includes 10,332 examples (48.7%) with images. After supplementing with images, the new dataset has 16,058 examples (75.7%) with images. The remaining 24.3% without images are mostly in the language science category, where relevant images couldn’t be found. The natural science category saw the most significant increase, nearly doubling the number of examples with images.

Both ARC-Easy and OpenDataEval are originally text-based QA datasets. In the new datasets,

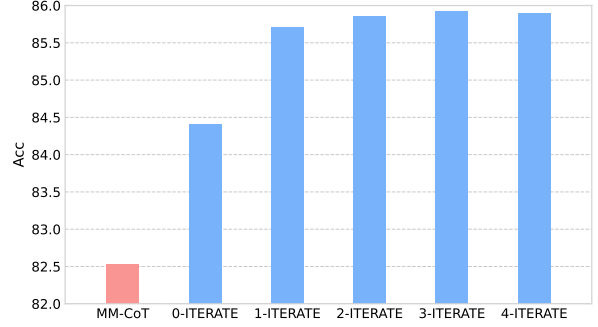


Figure 4: Ablation results of different iteration numbers on overall performance.

100% of examples now include images, meaning our method successfully found relevant images to assist in answering the text questions.

4.5 Main Results

Table 2 presents the results of our experiment with the ITERATE method. To ensure fairness, we compare it with MM-CoT_{Base}, using the same experimental parameters. ITERATE improved MM-CoT_{Base} by 3.39%, outperforming all baseline models. Notably, the TXT and NO categories within ScienceQA benefited significantly from paired images, enhancing the model’s cognitive abilities. The SOC category saw the largest accuracy boost, nearly 10%.

It is worth noting that, despite no changes in the LAN category examples, there was still a 1.8% uptick in accuracy. A credible rationale for this improvement can be found in the research by Lin et al. (Lin et al., 2023), which suggests that cross-modal understanding between different modalities (like image-text) in multimodal learning can not only enhance the performance of the model on multimodal tasks, but also potentially enhance its single-modal task performance.

The zero-shot performance improvements of ITERATE are shown in Tables 3 and 4. Our ITERATE method significantly enhanced LLaVA’s performance on these NLP tasks, demonstrating the value of the added image modality. LLaVA’s performance improved by approximately 5% and 7% across two datasets. Notably, LLaVA-13B with ITERATE outperformed the 33B LLaMA and the powerful Mistral-7B (matching the performance of LLaMA-33B with only 7B parameters), also surpassing LLMs with similar parameter counts.

Method	NAT	SOC	LAN	TXT	IMG	NO	Avg
MM-CoT _{Base}	82.24	83.13	82.64	80.84	76.15	85.64	82.53
MM-CoT _{Base} + ITERATE	83.88	92.91	84.45	82.65	82.05	86.90	85.92
w/o top-2 images select	82.80	92.52	83.01	81.57	80.97	85.96	84.87

Table 5: Ablation results on the ScienceQA of top-2 images selection module in ITERATE (%).

Method	ARC-E	ODEval
LLaVA-13B	78.0	64.8
LLaVA-13B + ITERATE	82.8	72.0
w/ shuffled images	74.9	66.6

Table 6: Ablation results of whether the images are correctly matched on ARC-Easy and OpenDataEval.

4.6 Ablation Analysis

Effect of Iteration Numbers In our ablation study on the ScienceQA dataset, we examined the impact of the ITERATE method and the number of iterations on model performance, as shown in Figure 4. Even with 0 iterations (0-ITERATE), the model’s accuracy increased from the baseline MM-CoT_{Base} model’s 82.53% to 84.41%, indicating that the image retrieval enhancement provided by ITERATE offers a certain performance advantage. Performance continued to improve with more iterations, reaching a peak accuracy of 85.92% at 3 iterations, indicating that iteration in ITERATE provides additional gains. We also analyze the impact of varying iteration numbers within ITERATE on the accuracy across six categories in ScienceQA, as shown in Appendix B.

Effect of top-2 selection module We conducted an ablation study on the top-2 selection module in ITERATE, used for selecting parent images for evolution. The number of iterations was set to three, matching the best performance setup. The control group used random sampling for parent images, with results shown in Table 5. Removing the top-2 selection module led to a 1.1% drop in overall performance, similar to ITERATE without iterations, highlighting the significance of the top-2 image selection module in enhancing the system’s overall efficiency. The NAT category showed the most significant performance decline, while the SOC category was less affected, indicating that SOC questions are less sensitive to image quality.

Effect of correct images pairing To prove that it is indeed the aligned content in the images that

aids the model in answering, rather than just the additional image information, we conducted corresponding ablation experiment where we shuffle the images for random pairing. The result as shown in Table 6, indicates that the performance improvement with shuffled image pairings is not significant, and there was even a decline, which emphasizes the correctness and effectiveness of our method.

5 Conclusions

In this paper, to improve the model’s performance on question-answering tasks, we introduce an additional visual modality (image) to enhance the model’s understanding and answering capabilities. Moreover, to obtain high-quality images, we introduce a cross-modal evolutionary method for image-text retrieval optimization, named ITERATE, inspired by evolutionary algorithms in reinforcement learning and the concept of the Visual Genome. Our experiments have demonstrated the helpfulness of additional modal information in enhancing the performance of purely NLP tasks, as well as the effectiveness of ITERATE in optimizing the quality of retrieved images. Our work takes an important step forward in improving the model’s performance on question-answering tasks by introducing high-quality new modalities, with ITERATE’s significant improvements in image retrieval and quality optimization highlighting the value of this approach and the potential of multimodal enhancements to task performance.

6 Limitations and Future Works

Our ITERATE method has proven effective across three QA datasets, and we now seek to validate it on more diverse tasks. However, ITERATE faces challenges; ablation studies show performance may decline after the third iteration, likely due to the increasing complexity of image-text matching. Future work could focus on selecting high-quality image pairs by enhancing the model’s understanding of task questions, rather than relying solely on image-text similarity.

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A Pseudocode of ITERATE

In Algorithm 1, initial keyword extraction operation, $Extr(\cdot)$, and evolutionary operator, $Evo(\cdot)$, are both conducted through few-shot learning in LLMs. The corresponding prompts are as follows:

- $Extr(\cdot)$: I have a question and its options now. Please extract a keyword for a noun or adjective from it and use them to search for images for data enhancement, making the QA results better. The following are some examples of keywords extraction:
Example 1: Question: {question of Example 1}, Options: {options of Example 1}, Keywords:{initial keyword of Example 1}.
Example 2: Question: {question of Example 2}, Options: {options of Example 2}, keywords:{initial keyword of Example 2}.
Now, give me the keywords based on the following information:
Question: {question of QUERY},
Options: {options of QUERY}
- $Evo(\cdot)$: Next, I will provide content descriptions of two images, which are obtained from a Bing search using the original keywords. I hope you can help me generate new keywords from the given two description sentences that encapsulate their differences. These keywords will be used for Bing image search to obtain higher quality images to assist in answering the question. The following is the example:
Original Keywords: {Original Keywords of Example}, Description 1: {content description of image-1}, Description 2: {content description of image-2}, New Keywords: {New Keywords of the Example}.
Now, give the new keywords based on the following information:
Original Keywords: {Original Keywords of QUERY}, Description 1: {content description of image-1}, Description 2: {content description of image-2}

B Ablation results within six categories examples of ScienceQA

Without any iterations, we find the SOC and IMG categories to have the most substantial performance gains, approximately 10% and 5%, respectively. However, for the other four categories, the performance of 0-ITERATE is only on par with the baseline. One possible reason is that the images retrieved without iteration optimization might be of poor quality, failing to effectively enhance performance, and even possibly harm the question-answering capability of the model. Consequently, after one single iteration, all these categories exhibit significant performance improvement, suggesting that this initial iteration plays a pivotal role in optimizing the quality of paired images. For instance, the accuracy increases by about 1.6% in NAT, TXT, and NO categories.

As the number of iterations increases to 2-3 times, the performance progressively stabilizes. This non-linear trend provides valuable reference for iteration strategies in practical applications, indicating that choosing the appropriate number of iterations is crucial for ensuring both enhancement and stability of model performance.

Algorithm 1 Pseudocode of ITERATE(x,n).

Require: x represents the input of sample t from task T , including question q and choices c , LLM as an evolutionary operator $Evo(\cdot)$, multimodal models denoted as M_{sel} and M_{extr} , Search engine denoted as $search(\cdot)$, empty image cluster IC

Ensure: Best Image img^*

```
1: Initial  $keywords = Extr(q, c)$  and  $imgs = search(keywords) \rightarrow IC$ 
2: for  $t = 1$  to  $n$  do
3:   Selection: select the best two images from the images cluster as parent.
4:      $similarity = M_{sel}(IC, q)$ 
5:      $ParentImgs = top2(similarity)$ 
6:   Reverse-Transcription: extract DNA described in natural language
7:      $captions = M_{extr}(ParentImgs)$ 
8:   Recombination: generate new keywords based on natural language DNA and question text
9:      $newKeywords = Evo(keywords, captions, q)$ 
10:  Update:  $ChildImgs = search(newKeywords)$  — replace  $\rightarrow IC$ 
11: end for
12: Return: the best image,  $img^*$ , matched from the cluster  $IC$ 
13:    $img^* = top1(M_{sel}(IC, q))$ 
```

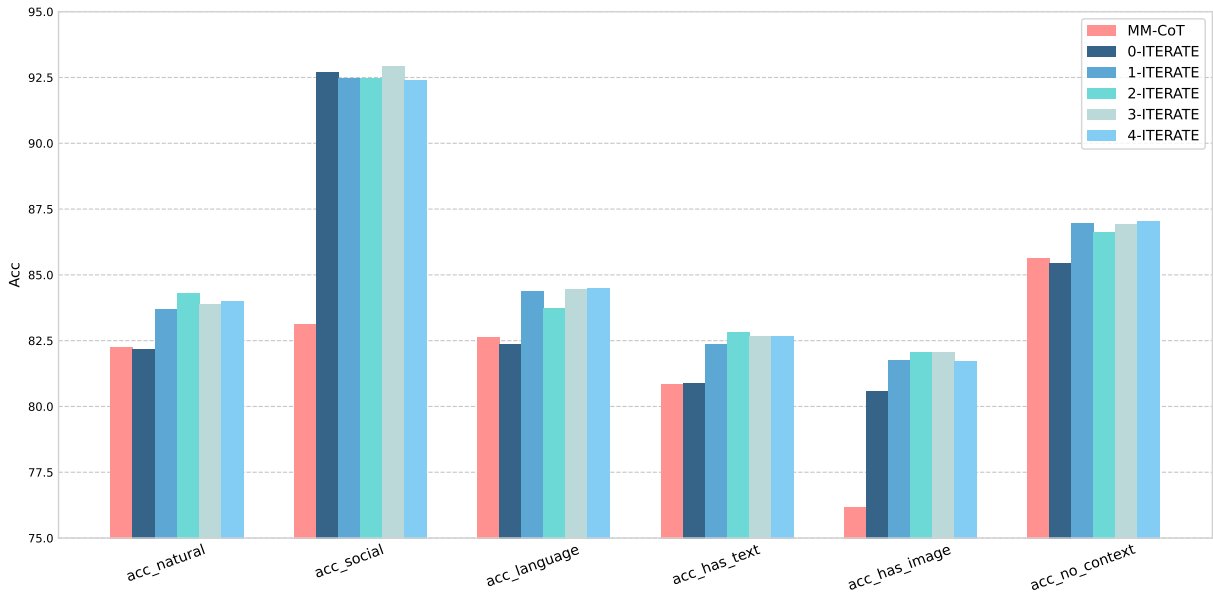


Figure 5: Ablation results of different iteration numbers on examples across six categories.